Lecture 3; Concentrations

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Concentrations and Absolute Numbers

Last time was mostly about sizes. Today we'll talk about concentrations and absolute numbers.

2 Homework

How's it going?

Want to mention that you should **list assumptions** you make

3 Reading the book

In the homework we ask you to read parts of the book. You all got a copy – I hope you'll read more than we assign.

One of the reasons we ask you to read is because the book is broken up into vignettes. These are short 2-3 pagers that ask and answer a quantitative question like 'how large are proteins?' For the final project you'll have to write a vignette, so reading will help you understand what it is.

Orientation: where we are in the course. Reflections on exercises: send clearly written docs, ideally as PDF. When you use numbers, supply a reference; we do not believe in "rumors." Reading vignettes should bolster understanding and show what is expected in final presentations.

4 mg / mL and volume–mass conversions

Very common to write 1 mg = 1 mL. What's the problem? 1 mL does not weigh 1 mg.

Volume–mass conversions are a common source of confusion:

$$1 \text{ mL} \approx 1 \text{ g} = 1 \text{ mkg}; \quad 1\ \mu\text{L} \approx 1 \text{ mg} = 1\ \mu\text{kg}; \quad 1 \text{ nL} \approx 1\ \mu\text{g} = 1 \text{ nkg}$$

$$1 \text{ pL} \approx 1 \text{ ng} = 1 \text{ pkg}; \quad 1 \text{ fL} \approx 1 \text{ pg} = 1 \text{ fkg}$$

Rule of Thumb: Living matter has ρ like water

mass density of water $\rho = M/V = 1$ kg/L

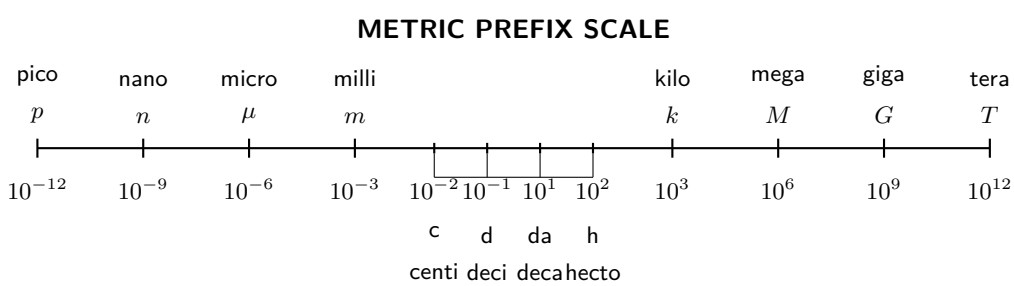

Figure 1: Metric prefixes. Made with the help of Claude.

Outline

- hw – assumptions
- book – reading
- units (mg / mL)
- sig digits
- 5 min
- 3 cells
- rule
- popn. NYC
- H^+ / cell
- 1-molecule conc
- DIY (break to calculate)
- rule of thumb
- Mammalian cell
- p53
- water
- H^+
- ping pong (break)
- why abs. numbers
- ants
- slides
- teaser – rates

5 Significant digits

My student James dropped into my office to talk. I told him "give me 5 minutes." After 50 s pass, you wouldn't say that we'll talk in 4min 10s, right?

This is too much precision for the situation. Another way of saying this: James knows the timing of our meeting to one sig fig. not two.

5.1 Three cells

If three HeLa cells together have a total volume of $10{,}000\ \mu\text{m}^3$, what should I report as the volume/cell? **What would you do?**

In papers you'll often see numbers like $3.426 \times 10^3\ \mu\text{m}^3$ per cell. People have their excel spreadsheets, so the arithmetic and report the exact answer.

Claim: You should write that the cell volume $V \approx 3000\ \mu\text{m}^3$ — *not* $3333.3\ \mu\text{m}^3$.

Lets talk about this.

Rule: In this course (and in most cases in biology), always report **1 significant digit** unless there is a special reason to report more. In intermediate calculations use 2 significant digits unless there is a special reason.

For this example you should report $V \approx 3 \times 10^3$. If you provide more significant digits ($3.33 \times 10^3\ \mu\text{m}^3$ per cell) it conveys a degree of accuracy that you don't have. You don't know the volumes of the individual cells, they may be very different from each other etc.

In your research: in your research, don't use more than two significant digits. If you use more, it's because you've quantified the accuracy of your measurement thoroughly.

Mention a case where you have an accurate quantification of 3 cells' volume. Or, alternatively, many V estimates for collections of n cells. In that case you might do a fuller analysis including error propagation to estimate the mean size (and standard deviation) of the single cell volume.

Several lessons to take here.

1. report the number
2. be honest with yourself about your measurement accuracy
3. the way you communicate a number gives a sense of the accuracy with which it is known/measured

5.2 Exception for differences of large numbers

What was the population change in NYC last year?

$$8.478 \times 10^6 - 8.391 \times 10^6 = 0.087 \times 10^6 \ \Rightarrow \text{ report as } \sim 0.1 \text{ million}$$

How many people is this? Staten Island has a population of 500,000. My mom lives in called Teaneck, NJ, which has a population of 40,000. So in 2024 NYC added $1/5$ a Staten Island or 2 Teanecks.

It's worth considering what the longer-term trends are. In Figure 2 you can see that there was fairly consistent growth in the NYC population until 2015. In 2019-2022 there was a steep decline (losing $\approx 400,000$ residents during COVID) from which we are now rebounding.

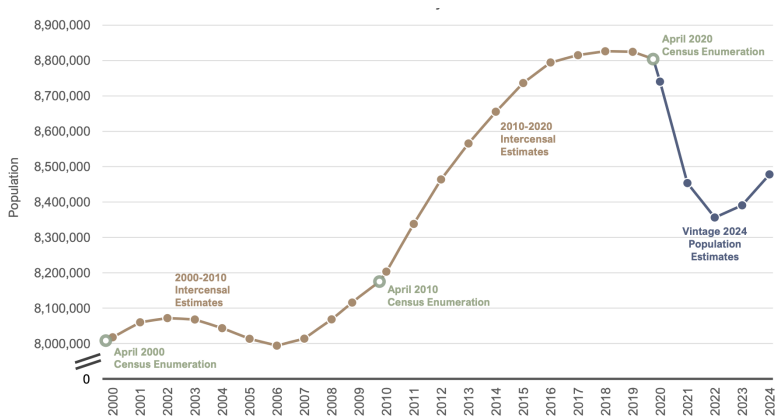

Figure 2: Changes in the NYC population since 2000. Source: NYC Department of City Planning, May 2025 Report

It's worth considering how accurately these numbers are known, and what they represent. Like many examples in biology, changes in population reflect the balance of migration, birth, and death. So we need to pay attention to what the numbers mean when we take their difference (derivative). We should also pay attention to how the numbers are measured – e.g., is it by counting or sampling or some other methods?

Practical example: Your calculation says the turn is 2.13 miles ahead. What should you actually say? "Drive about 2 miles north and then look for the sign."

Takeaway for cells: if you want to measure **changes** in cell size or shape, e.g., to measure growth, death, or morphogenesis, then you need greater measurement accuracy.

5.3 Difference between $\sim$ and $\approx$.

I just got an email that the 2027 RU graduate student stipend is $55,620 or $4,635 per month. You could write $\approx 55,000$ or $\approx f \times 10^4$ but not $\$\approx 10,000$ or $\$\approx 10^5$.

The symbol $\approx$ means "approximately equal to" meaning equal up to a factor of a few. The symbol $\sim$ means "same order of magnitude" or equal up to a factor of 10. So you could say that the stipend $\$\sim 10,000$, but it's better to convey more accuracy when you've got it.

You can read more about this and related issues in the book chapter titled The Order of Magnitude Biology Toolkit.

6 H^+ in a Cell

How many free H^+ / H_3O^+ ions are in a bacterial cell?

1. 10^2
2. 10^4
3. 10^6
4. 10^8

6.1 Quiz, Part 2:

If there is exactly **one molecule** in a bacterial cell, what is its concentration?

1. ~ 1 fM
2. ~ 1 pM
3. ~ 1 nM
4. ~ 1 μM
5. ~ 1 mM

(5 minutes for deliberations.)

My approach: you can always multiply or divide by 1.

We want to go to units of mols/L from molecules/cell. First lets get our volume units right.

$$\frac{1 \text{ molecule}}{\text{bacterial cell}} \times \frac{1\text{cell}}{1\mu\text{m}^3} \times \frac{1\mu\text{m}^3}{10^{-18}\text{m}^3} \times \frac{1\text{m}^3}{10^3\text{L}} = \frac{1 \text{ molecule}}{10^{-15}\text{L}}$$

We got L units. Now lets get moles.

$$\frac{1 \text{ molecule}}{10^{-15}\text{L}} \times \frac{1\text{mol}}{10^{24}\text{molecules}} = \frac{10^{-9}\text{mole}}{\text{L}} \approx 1 \text{ nM}$$

Notice how I rounded Avogadro's number to the nearest order-of-mag for simplicity/convenience.

Rule of Thumb: 1 molecule in a bacterial volume

1 molecule in a bacterial volume (e.g., *E. coli*) has a concentration of ≈ 1 nM.

6.2 1 molecule in a mammalian cell

Should the concentration be larger or smaller? Since the cell is larger (about 1000 times by colume) then the concentration should be smaller by about 1000x. That gives us 1 pM $= 10^{-12}M$.

Let's just check ourselves. According to [BNID 103725], HeLa cells have a typical volume of $f \times 10^3\mu\text{m}^3$. While these cells have a diameter of $f \times 10$ microbes, they are adherent and should not be thought of as spheres (see Fig. 3).

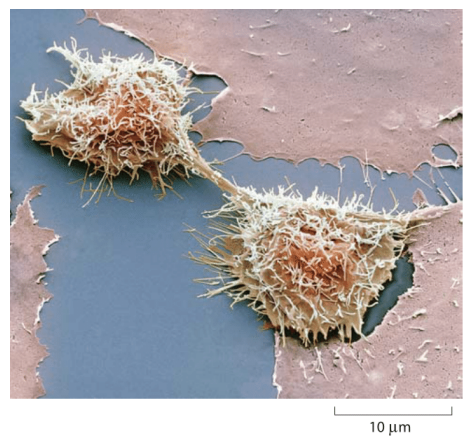

Figure 3: Dividing HeLa cells.

Of course it depends on what cell type exactly, but this is a reasonable starting point for many mammalian cell types. The book has a vignette on mammalian cell volumes for your reference.

Suppose we are thinking about yeast or green alga? What would the concentration be? More broadly, how does the single molecule concentration depend on r a spherical cell of radius r?

7 Signaling example: p53 in cells

If you ask people who study p53 (or other proteins) how many copies are in the cell, they often say "it depends." This doesn't satisfy us.

How many p53 molecules are there in a mammalian cell? $10^3, 10^5, 10^7, 10^9$, or 10^{11}?

People in the field have much better intuition about concentrations than absolute numbers. p53 (and many other signalling proteins) is present at ~ 1 μM in a mammalian cell, corresponding to $\sim 10^6$ copies per cell.

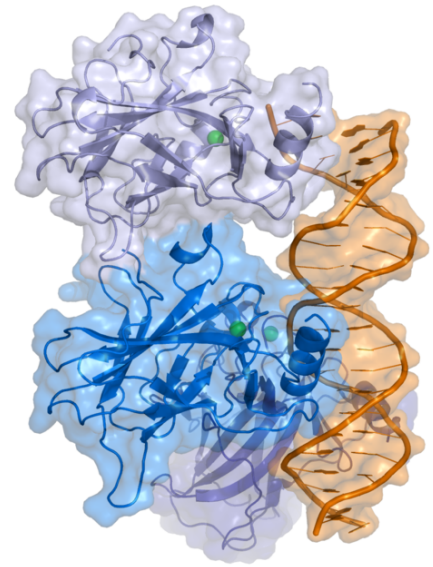

Figure 4: p53 in complex with cognate DNA.

7.1 Why does the absolute number p53/cell matter?

What does p53 do? It is involved in a complex signaling pathway, but it ultimately binds DNA to effectuate a transcriptional response. There are $\approx 10^3$ p53 binding sites in the genome, so the estimated concentration is $\sim 10^2 - 10^3$ fold in excess of the number of binding sites.

Did this comparison sharpen your thinking? What can you infer?

TF	Copies per cell	Approx. conc.
LacI	10–100	sub-nM
Gal4	$\sim 10^2$	~nM
Bicoid	$\sim 10^4$	~ 10 nM
p53	10^4–10^5	$\sim \mu$M

7.2 Other reasons why absolute numbers matter

Knowing absolute numbers (not just concentrations or ratios) matters for three main reasons:

1. **Stochasticity** — small copy numbers mean large fluctuations.
2. **Intuition** — numbers anchor reasoning and reveal when something is surprising.
3. **Memorability** — round-number rules of thumb are reusable across contexts.

7.3 How many water molecules per cell?

You can remember the rule of thumb that the concentration of water ≈ 50 M in aqueous settings.[1] Alternatively, you can derive it quickly from the fact that 1 L = 1 kg and water has a molar mass of $16 + 2 \approx 20$.

$$\frac{1000\text{g}}{\text{L}} \times \frac{1\text{mol}}{20\text{ g}} = 50 \text{ mol/L}$$

Rule of Thumb: Water Concentration

The water concentration $[H_2O] \approx 50$ M.

Let's calculate for a bacterium first. If the concentration is 50 M and 10^{-9} M is 1 molecule, then we have $\approx 50 \times 10^9 = 5 \times 10^{10}$ water molecules per bacterial cell volume. This is like number of people on Earth (roughly 5x more). [2]

The mammalian cell is about 1000x larger, so:

[1] The concentration of water in pure water is 55 M.

[2] Try to correct the water concentration for the other stuff, which makes up $\approx 30\%$ of cell mass. Convince yourself that this doesn't change the water concentration much, i.e., its not even a factor of 2 change.

- Bacterial cell: $[H_2O] \sim 10^{10}$–10^{11} molecules.
- Mammalian cell: $\sim 10^{13}$–10^{14} molecules.

Returning to the quiz — H^+ in a bacterial cell:

We have our rule of thumb (1 molecule = 1 nM) but we need a concentration. We get this from pH ≈ 7.

pH is defined as $-\log_{10}\left([H^+]\right)$.[3] So the concentration $[H^+] \approx 10^{-7}$ M. What's the absolute number of H+ in a bacterial cell volume.

$$[H^+] \approx 10^{-7}\ \text{M} = 100\ \text{nM} \ \Rightarrow \sim 100 \text{ molecules per bacterial cell}$$

We did the math and we got a number. 1) I hope I convinced you that the rules of thumb are useful. We used one number and derived a lot from it. 2) Is the H+/bacterium number surprising?

Seems like a low number, especially since we know that H+ are involved in really important processes like energetics (pumping across membranes) and catalysis (enzyme active site mechanisms). How can we resolve this apparent conflict?

- **Buffering:** The buffer capacity B is defined as the amount of strong acid required to change pH by one unit; typically $B \sim 10$–100 mM. (See ionic composition and pK_a of proteins and phosphates.)
- **Lifetime:** The lifetime of an H^+ ion in water is $\sim 10^{-12}$ s. Every ion "visits" every water molecule in the cell ~ 10 times per second; with ~ 100 ions per cell, every water molecule is transiently converted to H_3O^+ roughly 10^3 times per second.
- Continuous annihilation and formation.
- **Diffusion:** of H+ in water is very fast, see Wikipedia on the Grotthuss mechanism.

In sum: all of us knew pH 7, which is a concentration. Once we converted to absolute numbers, it raised all sorts of questions at the interface of biology and chemistry.

*From a correspondence with **Clive Bagshaw** (UC Santa Cruz / Emeritus, University of Leicester):*

> From Meiboom's NMR data, the rate constant for the exchange reaction $H_2O + H_3O^+ \rightleftharpoons H_3O^+ + H_2O$ is 10^{10} $M^{-1}s^{-1}$. At $[H_2O] = 55$ M this gives a pseudo-first-order rate constant of 5×10^{11} s^{-1} and a mean H_3O^+ lifetime of 2×10^{-12} s. Since H_2O is 5×10^8 times more abundant than H_3O^+ at pH 7, each water molecule on average experiences the reaction on a timescale of $\sim 10^{-3}$ s.

Further reading: Aminov et al. (2019), "Intact carbonic acid is a viable protonating agent for biological bases," *PNAS*.

How does a water molecule feel inside a cell?

Scale up: let a water molecule be the size of a bee (~ 0.1 g, volume ~ 100 mm$^3 \approx 10^{-7}$ m^3). There are $\sim 10^{10}$ water molecules in a bacterial cell, so the total scaled volume is:

$$10^{10} \times 10^{-7}\ \text{m}^3 = 10^3\ \text{m}^3 \approx \text{a few lecture rooms}$$

[3] H in pH is for H+ and p is for power.

So: *about 100 bees in a room just larger than ours, doing random walks — but also continuously appearing and disappearing.*

8 1 in a billion, linear vs 3D

Marty Supreme challenge: if I line up ping-pong balls along the equator, how many balls are there?

I only remember the **radius of the Earth**, which is ≈ 6000 km $= 6 \times 10^6$ m. This gives a circumference of $2\pi r \approx 40 \times 10^6 \approx f \times 10^7$ m.

A ping-pong ball is a few cm in diameter (regulation diameter is 40 mm = 4 cm), or 10^-2 m. So I can line up more than 10^9 ping-pong balls along the equator. If one of them is orange (as in Marty Supreme) then its *one in a billion*.

Tells you something about how big is a billion.

Now in volumetric units I have a special green marble that I toss into an Olympic swimming pool full of blue marbles. What is the concentration of the single green marble?

Olympic swimming pool is 50 m x 25 m x 2 m = 2,500 m^3. 1 m^3 = 1000 L, so 2.5×10^6 L. If I fill this pool with marbles with 1 cm radius (V $= \frac{4}{3}\pi \approx 4$ cm^3. How many marbles fit in the pool?

$$\frac{1 \text{ marble}}{f \text{ cm}^3} \times \frac{1 \text{ cm}^3}{10^{-3} \text{ L}} \times \frac{f \times 10^6 \text{ L}}{\text{pool}} \approx \frac{10^9 \text{marbles}}{\text{pool}}$$

Our green marble is **one in a billion**.

8.1 Ants and very big numbers

Recently a paper claimed that there are 20 quadrillion ants in the world.[4] The series is a million, billion, trillion, quadrillion, so quadrillion is 10^15.

Claim: There are 2×10^{16} ants in the world. Is this plausible?

We just talked about how we have better intuition for absolute numbers than concentrations. But we also lack intuition for really large absolute numbers (like the ping-pong balls). To think about this number we can bring it back to human scale — normalize it to a 1 m^2 area (1 m $\approx 3 - 4$ ft).

Sanity check by areal density:

Surface area of Earth $\approx 4\pi \times (6000 \text{ km})^2 \approx 10 \times (6 \times 10^6 \text{m})^2 \approx f \times 10^{14} \text{m}^2$

Of this, only $\approx \frac{1}{3}$ is land, giving $\approx 10^{14}$m^2 total.

$$\frac{f \times 10^{16} \text{ ants}}{10^{14} \text{ m}^2} \approx f \times 10^2 \ \frac{\text{ants}}{\text{m}^2}$$

About 100 ants per square meter of Earth's surface. Remember that about — **plausible!**. If we look right here, we see no ants, but if we look at a colony we'll see much more.

[4] See Schultheiss et al. PNAS 2022.

Note: we achieved a sanity check by normalizing a large absolute number to an intuitive density.

Some useful rules: US is 10 million square kilometers. China and Canada: 10 million. Russia: 20 million. All of Europe: 10 million. Sum it up: about 100 million km^2 land area on Earth (a more exact number is 150×10^6 km^2).

Rule of Thumb: Land Area of Earth

Land area of Earth $\approx$ 100 million km^2 $= 10^8$ km^2

8.2 Slides on concentration and absolute numbers

Things to note on the slides.

- protein and mRNA abundances are measured in log2
- concentration x charge equal inside and outside: electroneutrality.
- protein is the dominant constituent of biomass
- the total metabolite concentrate $f \times 10^2$ mM
- glutamate is the most abundant metabolite: why?!
- $f \times 10^6$ proteins/$\mu\mathrm{m}^3$ $\rightarrow\approx f$ mM total protein concentration.
- number of proteins/cell for different cell types

Most abundant proteins:

- *Cell:* evenly divided between metabolism and central dogma.[5]
- At a finer scale: central metabolism and translation dominate.
- Explain Voronoi treemaps – amount $\propto$ area.
- Many more unmapped protein counts in human.
- *Whole body:* Collagen (25–35% of total protein).
- *Biosphere:* rubisco.

"Signalling has great PR." What would happen if signaling proteins had much higher concentrations?

Highly expressed proteins dominate: in yeast the top 10 proteins account for $> 10^5$ ppm or $\approx 10\%$ of the total protein counts (see PaxDB for yeast). Similar results arise in *E. coli* and human cells, though *the top 10 proteins are different* in each organism.

Metabolites are more concentrated than proteins by 100x or so. Does this make sense? Certain key metabolites are much more concentrated than others. Do you have ideas as to why?

Project idea: Choose one protein and discuss *why* its concentration is what it is — not merely that it is 10-fold lower or higher than average.

9 Teasers

What is faster in bacteria — transcription or translation?

1. Translation
2. Transcription
3. Roughly the same

Teaser 2: What is the characteristic power consumption of a human?

[5] Yeast data.

1. 1 W
2. 10 W
3. 100 W
4. 1000 W

Teaser 3: What is the characteristic power consumption of a $1000\ \mu$m^3 human cell?

1. 10^{-9} W
2. 10^{-12} W
3. 10^{-15} W
4. 10^{-18} W

What is the total weight of bacteria in our body?

1. 0.1 kg
2. 0.3 kg
3. 1 kg
4. 3 kg
5. 10 kg

10 Reminders

Assignment, continue bringing ideas for the final project.